

Mathematical Modeling of K-pop Fan Growth

A Case Study of BOYNEXTDOOR

Yuanyuan Tang

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1. Introduction

1.1 Rationale

The pop music industry in South Korea has always had a relatively significant influence in Asia, with a massive fan base. Some reputable and influential entertainment agencies continuously launch new idol groups, and these groups can quickly accumulate a large number of fans during their debut period and gradually gain more fans in future comebacks. During this process, the acquisition of fans is influenced by various factors, such as the quality of groups' albums, the members' stage performing capacity, and their personalities and interactions as portrayed in the variety shows.

As someone deeply influenced by K-pop culture, I am intrigued by this dynamic model of fan base expansion. Therefore, I began to wonder whether mathematical models could be used to simulate and predict the growth of fan communities, especially for BOYNEXTDOOR, a relatively young idol group that I have been a fan of. By applying mathematical models, the growth trend of this group's fanbase can be revealed, and changes in the number of fans over a given period in the future can be predicted. This kind of prediction has significant practical implications for their entertainment company. For example, based on the forecast data, their company HYBE can plan the quantity of physical album releases and devise production schedules for merchandise for their next comeback, in order to optimize resource allocation and thereby maximize profits.

1.2 Aim

The aim of this investigation was to explore and evaluate potential mathematical models that can effectively simulate and predict the fan growth of the K-pop group BOYNEXTDOOR. In this study, I chose to use YouTube subscriber count as a representative indicator of BOYNEXTDOOR's fanbase. This is because, compared to platforms like Instagram and TikTok, which tend to attract casual viewers more easily through images and short videos, YouTube subscribers are typically dedicated fans who consistently follow the group's music videos, behind-the-scenes content, and other long-form materials. After collecting monthly subscriber data from BOYNEXTDOOR's debut to the present as the basis for analysis (Social Blade, 2025), I applied it to four different modeling approaches: polynomial regression, exponential growth, logistic growth, and the Bass Diffusion Model. By comparing the predicted values of each model with the actual data, my goal was to

determine which model can best represent the growth of BOYNEXTDOOR's fanbase and may be useful for future forecasting and industry decision-making.

1.3 Introduction to Terms and Data

For the sake of clarity, I will be using the following denotations throughout my Math IA:

- S_n represents the YouTube subscriber count, used as a measure of the fan base
- S_0 represents the initial subscriber count at the time of BOYNEXTDOOR's debut
- t_n represents the month since debut
- R represents the growth rate, used in both the exponential and logistic growth models
- C represents the carrying capacity, which refers to the maximum potential subscriber count
- p represents the coefficient of innovation (external influence)
- q represents the coefficient of imitation (internal influence)
- m represents the total market potential

t_n	S_n	t_n	S_n	t_n	S_n
0	6350	7	595000	14	1000000
1	80400	8	632000	15	1070000
2	242000	9	668000	16	1130000
3	358000	10	700000	17	1220000
4	473000	11	734000	18	1280000
5	538000	12	836000	19	1320000
6	571000	13	932000	20	1380000

Table 1: Monthly YouTube Subscriber Count (S_n) of BOYNEXTDOOR Over 20 Months Since Debut

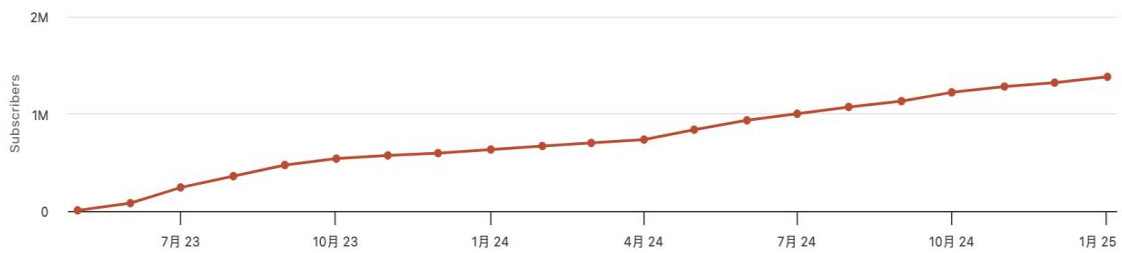


Figure 1: Monthly YouTube Subscriber Growth for BOYNEXTDOOR Since Debut (Social Blade, 2025)

2. Application of Mathematics Models

2.1 Polynomial Regression

To accurately model the fan growth of BOYNEXTDOOR, I will first employ polynomial regression to determine the best-fitting curve for the dataset. While simple linear regression assumes a constant rate of increase, fan growth in the real world is influenced by multiple factors like promotions and word-of-mouth effect, and usually occurs as a non-linear pattern. Therefore, polynomial regression appears to be a powerful method for modeling, as its incorporation of higher-degree terms allows the curve bend to follow the data.

The general form of a polynomial regression model of degree n is:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \cdots + \beta_{n-1} x^{n-1} + \beta_n x^n + \varepsilon$$

where:

- $\beta_0, \beta_1, \dots, \beta_n$: polynomial coefficients to be determined
- ε : error term (residual)

From the point view of estimation, since the regression function is linear in terms of the unknown parameters $\beta_0, \beta_1, \dots$, polynomial regression can be simplified to a system of linear equations (GeeksforGeeks, 2018). Therefore, we can express it in matrix form in terms of a design matrix $\mathbf{X}$, a response vector y , a parameter vector β , and a random error vector ε :

$$\vec{y} = \mathbf{X}\vec{\beta} + \vec{\varepsilon}$$

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_m x_i^m + \varepsilon_i (i = 1, 2, \dots, n)$$

which can also be written as:

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^m \\ 1 & x_2 & x_2^2 & \cdots & x_2^m \\ 1 & x_3 & x_3^2 & \cdots & x_3^m \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^m \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_m \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

The goal is to determine β (the polynomial coefficients) that minimize the total squared error, which is also known as the sum of squared residuals, using the Ordinary Least Squares Method (Montgomery et al., 2021).

The squared error in matrix form is given by:

$$S = (\mathbf{y} - \mathbf{X}\beta)^T (\mathbf{y} - \mathbf{X}\beta)$$

By expanding the quadratic form,

$$S = \mathbf{y}^T \mathbf{y} - 2\boldsymbol{\beta}^T \mathbf{X}^T \mathbf{y} + \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{X} \boldsymbol{\beta}$$

To minimize S , we differentiate it with respect to $\boldsymbol{\beta}$:

Set the derivative equal to zero to find the minimum:

$$\frac{dS}{d\boldsymbol{\beta}} = -2\mathbf{X}^T \mathbf{y} + 2\mathbf{X}^T \mathbf{X} \boldsymbol{\beta}$$

Set the derivative equal to zero to find the minimum:

$$-2\mathbf{X}^T \mathbf{y} + 2\mathbf{X}^T \mathbf{X} \boldsymbol{\beta} = 0$$

$$\mathbf{X}^T \mathbf{X} \boldsymbol{\beta} = \mathbf{X}^T \mathbf{y}$$

$$\boldsymbol{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

This is the normal equation, which we solve for the optimal polynomial coefficients $\boldsymbol{\beta}$.

In polynomial regression, as the polynomial degree increases, the resulting model will be closer to the variation in the original data and more flexibly fit the nonlinear features. However, this also raises the risk of overfitting, where the model captures irregular fluctuations rather than the actual underlying trend. Therefore, choosing an appropriate polynomial degree is essential for constructing a model.

With the dataset, I chose to construct the Vandermonde matrix for degree 7.

$$\begin{bmatrix} 6350 \\ 80400 \\ 242000 \\ \vdots \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 16 & 32 & 64 & 128 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix} \cdot \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \\ \beta_6 \\ \beta_7 \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \vdots \end{bmatrix}$$

To compute the optimal coefficients, I used a matrix calculator (Desmos Matrix Tool) and this allowed me calculate the inverse and dot products without manual error.

The resulting coefficients (rounded to 2 decimal places) were:

$$\begin{aligned} \beta_0 &= 4443.20 & \beta_4 &= 2965.52 \\ \beta_1 &= 22960.44 & \beta_5 &= -182.75 \\ \beta_2 &= 82507.86 & \beta_6 &= 5.59 \\ \beta_3 &= -24073.28 & \beta_7 &= -0.068 \end{aligned}$$

The final polynomial model for BOYNEXTDOOR's fan growth is:

$$y = -0.068x^7 + 5.59x^6 - 182.75x^5 + 2965.52x^4 - 24073.28x^3 + 82507.86x^2 + 22960.44x + 4443.20$$

To visualize the accuracy of this model, I plotted the modeled values against the original data and the curve aligned closely with the actual fan counts across all 21 months.

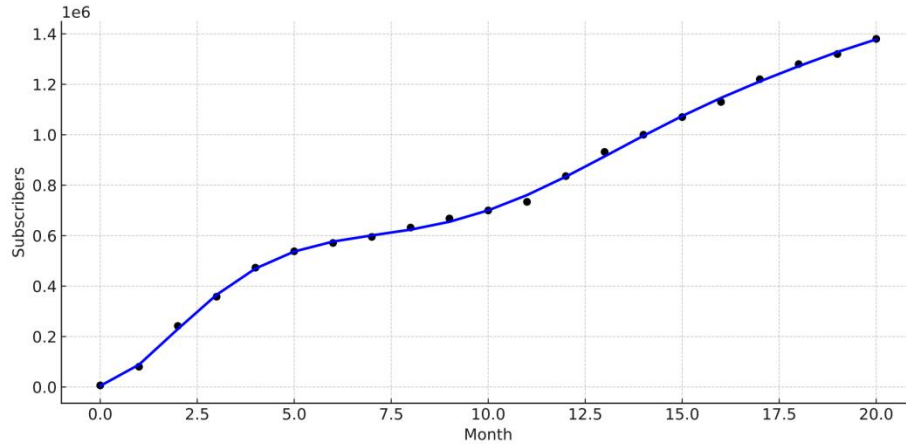


Figure 2: Comparison Between Actual Subscriber Data and 7th-Degree Polynomial Regression Model

Meanwhile, to evaluate the performance of the model numerically, I calculated the coefficient of determination R^2 , which measures the proportion of variance in the observed values explained by the model (MathWorks, n.d.).

$$R^2 = 1 - \frac{\sum (S_{\text{actual}} - S_{\text{model}})^2}{\sum (S_{\text{actual}} - \bar{S})^2}$$

$$R^2 \approx 0.9994$$

A value of R^2 close to 1 indicates a strong model fit. For my 7th-degree polynomial model, I obtained a value of 0.9994, which suggests that this model explains more than 99.9% of the variance in the actual fan data, indicating an exceptionally good fit.

As the first model I explored, polynomial regression demonstrated strong numerical fitting ability, as it precisely captured small fluctuations in the data. However, it is purely data-driven and lacks interpretability in the real-world context of fan growth, since the coefficients do not represent any meaningful social or marketing behavior. Moreover, although the polynomial model accurately simulates the data within the timeframe we have, it tends to behave unpredictably outside of that range. For instance, after the 20th month, the curve may rise or fall rapidly in a way that contradicts realistic fan growth trends. This limitation reveals that while polynomial regression is mathematically convenient, it functions more as a data-fitting tool rather than an explanatory or predictive model. Therefore, when applied to modeling BOYNEXTDOOR's fan growth, it cannot offer practical value for their company for resource planning or strategic decision-making.

2.2 Population Growth Model

2.2.1 Exponential Model

Due to the foundational role of the exponential model in mathematics and its widespread application across various disciplines, I have chosen it as my second model to attempt. In 1798, Thomas Malthus introduced the concept of exponential growth in his seminal work, *An Essay on the Principle of Population*. It is argued that the number of individuals in a population grows at a constant rate over equal time intervals in the absence of limiting factors. Therefore, as the population size expands, the rate of population growth becomes faster and faster, resulting in an exponential increase. In an industry like K-pop, where fan growth is largely relied on information dissemination, the exponential model may be equally applicable. The model is given by:

$$\frac{dS_n}{dt_n} = RS_n$$

Here, R is a constant representing the growth rate of active fans (subscriber count). In this model, we assume that the rate of change is directly proportional to the fans population, we multiply S_n with R . This differential equation can be solved by applying separable variables. After grouping like terms together, we integrate both sides with limits t_0 to t_n , representing the time period over which we model the fan population growth:

$$\begin{aligned}\frac{dS_n}{S_n} &= R dt_n \\ \int_{S_0}^{S_n} \frac{dS_n}{S_n} &= \int_{t_0}^{t_n} R dt_n \\ [\ln |S_n|]_{S_0}^{S_n} &= [Rt_n]_{t_0}^{t_n} \\ \ln |S_n| - \ln |S_0| &= Rt_n - Rt_0 \\ \ln \left| \frac{S_n}{S_0} \right| &= Rt_n - Rt_0\end{aligned}$$

As we assume that the subscriber count is always increasing, $S_n \geq S_0$, therefore $\frac{S_n}{S_0} \geq 1$. The equation can be written without the modulus sign:

$$\begin{aligned}\ln \left(\frac{S_n}{S_0} \right) &= Rt_n - Rt_0 \\ \frac{S_n}{S_0} &= e^{Rt_n - Rt_0}\end{aligned}$$

As $t_0 = 0$ while $S_0 > 0$,

$$S_n = S_0 e^{Rt_n}$$

To find R , I then calculated the percentage change in the fan population for each month using the following equation:

$$R = \frac{\text{subscriber count in the current month} - \text{subscriber count in the previous month}}{\text{subscriber count in the previous month}} \times 100\%$$

t_n	S_n	% change	t_n	S_n	% change	t_n	S_n	% change
0	6350	0.00	7	595000	4.20	14	1000000	7.30
1	80400	1166.14	8	632000	6.22	15	1070000	7.00
2	242000	201.00	9	668000	5.70	16	1130000	5.61
3	358000	47.93	10	700000	4.79	17	1220000	7.96
4	473000	32.12	11	734000	4.86	18	1280000	4.92
5	538000	13.74	12	836000	13.90	19	1320000	3.13
6	571000	6.13	13	932000	11.48	20	1380000	4.55

Table 2: The Percentage Change for Monthly Subscriber Count for BOYNEXTDOOR

To find the value of R , I could use both mean and median percentage change. However, events such as new album releases and comebacks could trigger sudden, sharp increase in fan numbers that act as outliers. These atypical increases could significantly influence the mean percentage change. In this case, the median percentage change is less sensitive to those extreme values, such as 1166.14% and 201.00%, because it represents the middle value of the data set. Therefore, I chose to use the median value in my model, which turns out to be 6.61% :

$$\text{Median} = \frac{6.22\% + 7.00\%}{2} \approx 6.61\%$$

Expressed as a multiplier, the value of R is 0.0661 (3.s.f), giving us our first model:

$$S_n = 6350e^{0.0661t_n}$$

In order to test whether or not this mathematical model gives accurate results, I compared the graph of my obtained model to the line chart of the real data.

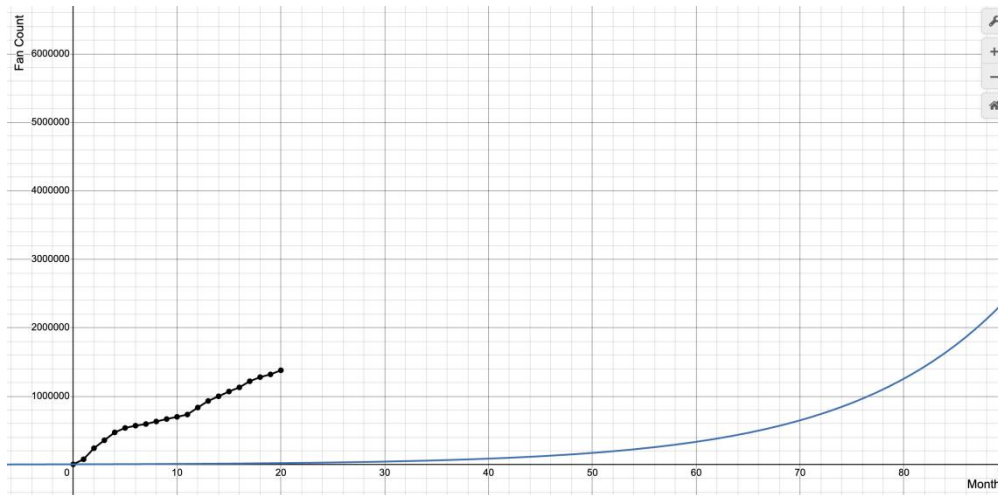


Figure 3: Comparison Between Actual Subscriber Data and the Exponential Model

As shown in the figure, the model I applied here demonstrates a poor correlation with the actual fan data of BOYNEXTDOOR. The exponential curve remained nearly flat in the first few months, and only began to rise significantly after the 50th month, which obviously does not reflect the realistic trend of fan growth. This misalignment is likely due to the fact that the core assumption of the exponential model — a constant growth rate — is inconsistent with the fluctuating nature of fan engagement in reality. In addition, I used the median to estimate the growth rate R , which may have suppressed the impact of the abnormal period, especially the rapid fan surge after the group's debut, leading to a severe underestimation of the promotions-driven early growth. In fact, K-pop fan growth cannot follow a purely exponential trend, as the growth rate often slows down during non-comeback periods due to reduced media exposure. As a result, the exponential model is not a good choice as it cannot capture the dynamics of the fan growth in K-pop industry, and therefore does not provide practical value in analysis and prediction.

2.2.2 Logistic Model

Like the exponential model, the logistic model is also a commonly used tool, especially in the context of population modeling. As the exponential model was being developed, the Belgian mathematician Pierre-François Verhulst observed that, due to the finite availability of resources such as food, space, and other environmental factors, population growth will eventually encounter

limitations. In 1838, he proposed the logistic model, which incorporates a carrying capacity, the maximum population level that an environment can sustain. In the K-pop industry, the number of fans a group can attract is constrained by factors such as market saturation, competition from other artists, and shifts in public interest. When the fan count approaches the maximum number that its market can sustain, the rate of fan growth will slow down. The model is given by:

$$\frac{dS_n}{dt_n} = RS_n \left(1 - \frac{S_n}{C}\right)$$

As a kind of generalization of the equation for exponential growth (Bacaër, 2011), logistic growth model introduced a multiplier with the constant C , which is the carrying capacity. When the value of S_n is small, the multiplier tends to close to 1, therefore the growth curve mimics the exponential model in the initial stages. As S_n increases, the value of the multiplier approaches 0, so the growth curve tends to C . Similar to the exponential model, this differential equation can be solved by separable variables:

$$\begin{aligned} \frac{dS_n}{S_n \left(1 - \frac{S_n}{C}\right)} &= R dt_n \\ \int \frac{dS_n}{S_n \left(1 - \frac{S_n}{C}\right)} &= \int R dt_n \end{aligned}$$

To integrate the left-hand side, I expressed it in terms of partial fraction:

$$\begin{aligned} LHS &= dS_n \times \frac{1}{S_n \left(1 - \frac{S_n}{C}\right)} = dS_n \times \frac{C}{S_n(C - S_n)} \\ \frac{C}{S_n(C - S_n)} &= \frac{A}{S_n} + \frac{B}{C - S_n} \\ C &= A(C - S_n) + B(S_n) \end{aligned}$$

To solve the partial fraction, I proceeded to substitute S_n with C ,

$$C = A(C - C) + B(C)$$

$$C = B(C)$$

$$B = \frac{C}{C} = 1$$

$$\text{now } C = A(C - S_n) + S_n$$

$$C - S_n = A(C - S_n)$$

$$A = 1$$

The original partial fraction can be rewritten as

$$\frac{C}{S_n(C - S_n)} = \frac{1}{S_n} + \frac{1}{C - S_n}$$

Now the equation can be integrated separately:

$$\begin{aligned}
\int \left(\frac{1}{S_n} + \frac{1}{C - S_n} \right) dS_n &= \int R dt_n \\
\int \frac{1}{S_n} dS_n + \int \frac{1}{C - S_n} dS_n &= \int R dt_n \\
\ln |S_n| - \ln |C - S_n| &= Rt_n + K \\
\ln |C - S_n| - \ln |S_n| &= -Rt_n - K \\
\ln \left| \frac{C - S_n}{S_n} \right| &= -Rt_n - K \\
\left| \frac{C - S_n}{S_n} \right| &= e^{-Rt_n - K} \\
\left| \frac{C - S_n}{S_n} \right| &= e^{-Rt_n} \cdot e^{-K}
\end{aligned}$$

Here, we substitute A with $\pm e^{-k}$,

$$\begin{aligned}
\frac{C - S_n}{S_n} &= Ae^{-Rt_n} \\
S_n Ae^{-Rt_n} + S_n &= C \\
S_n (Ae^{-Rt_n} + 1) &= C
\end{aligned}$$

When substituting S_n with S_0 and t_n with t_0 , the expression leaves A as a constant:

$$A = \frac{C - S_0}{S_0}$$

Therefore, we can get the equation in terms of S_n :

$$S_n = \frac{C}{Ae^{-Rt_n} + 1} \quad \text{where } A = \frac{C - S_0}{S_0}$$

In order to model the population growth of a K-pop group, I had to estimate its market carrying capacity, which was challenging. Unlike the total K-pop fanbase, the fan population of each group is constrained by factors such as the group's style and the unique characteristics of its members. As a result, the market carrying capacity C for a specific group like BOYNEXTDOOR cannot be directly obtained from official data through research. Therefore, to estimate C , I referred to the subscriber counts of several K-pop male groups that have been active for a few years. These groups, like the one being analyzed, have a specific audience due to their style, rather than a massive global influence like BTS, which could help provide a more realistic estimation. For example, SEVENTEEN, ENHYPEN, and TOMORROW X TOGETHER have a total YOUTUBE subscriber count of 13.8 million, 11.2 million, and 12.5 million respectively. Therefore, I would estimate the

carrying capacity C for BOYNEXTDOOR will be 15.0 million fans. As such, A can now be written as:

$$A = \frac{15000000 - 6350}{6350} = 2361.20 \text{ (2.s.f.)}$$

Adopt the R value calculated previously, we can have our logistic growth model for this group:

$$S_n = \frac{15000000}{1 + 2361.20 e^{-0.0661t_n}}$$

As before, I plotted the model to test if it was well-fitted.

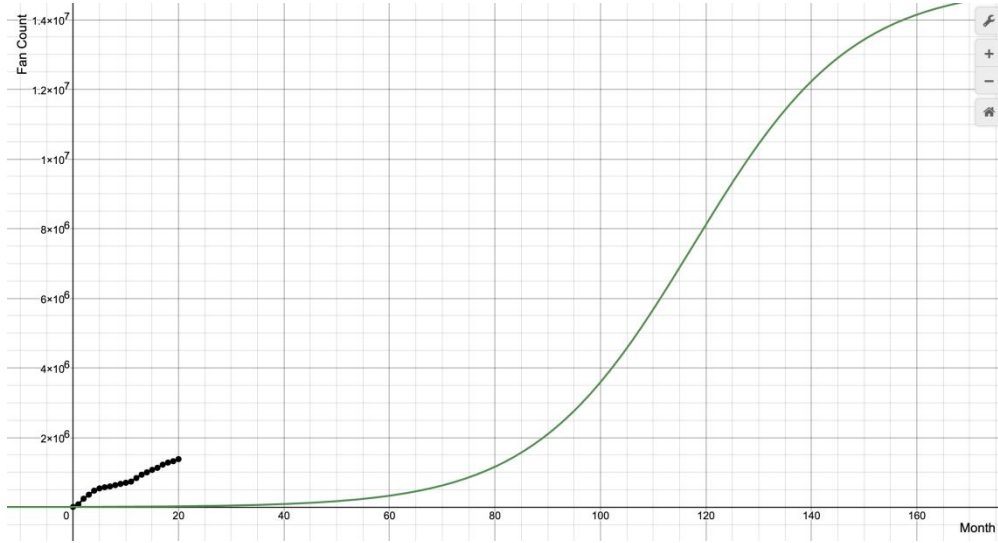


Figure 4: Comparison Between Actual Subscriber Data and the Logistic Model

As seen in the figure, I once again obtained a poorly correlated model. Although the logistic curve introduces a carrying capacity C and eventually levels off, the early stage of the model behaves almost identically to an exponential function since S_n is small during the debut period. Therefore, the limitation of this model similarly comes from its failure to account for the explosive growth in fan count during the group's debut period when calculating the growth rate.

Despite its introduction of C helping add realism to long-term projections, for newly debuted groups like BOYNEXTDOOR, the logistic model still cannot reflect the rapid accumulation of fans driven by media promotion in the early stage. As with the exponential model, it oversimplifies the adoption process, as it ignores the influence of external factors on the growth rate, and is thus not suitable for modeling or predicting the dynamic fan growth in the K-pop industry.

2.3 Bass Diffusion Model

After the failure of the population growth model, I began to reconsider the mechanism behind the increase in fan count. As K-pop idol groups are essentially “company-made products” — brought into the public sphere by publicity, packaging, and marketing operations — I believed that the growth in fanbase would be more aligned with product diffusion logic. Therefore, I decided to explore the Bass Diffusion Model, a mathematical model proposed by Frank Bass in 1969 for forecasting the adoption of new products within a population (Bass, 2004).

This model classifies adopters into two groups: innovators, who act early due to external influences such as advertising and promotion, and imitators, who adopt later due to the influence of existing adopters. The fan growth of K-pop idol groups often follows a similar pattern — it begins with early growth driven by core fans from pre-debut marketing, and then expands through mass adoption under the influence of social media.

The Bass model is described by the following differential equation:

$$\frac{dS(t)}{dt} = \left[p + q \cdot \frac{S(t)}{m} \right] \cdot [m - S(t)]$$

Mathematically, the basic Bass diffusion is a Riccati equation with constant coefficients, equivalent to Verhulst–Pearl logistic growth. Differently, it introduces interaction between adopters and potential adopters, with the expression $p + q \cdot \frac{S(t)}{m}$ capturing the combined influence of innovation and imitation. As $S(t)$ increases over time, the imitation component grows, thereby accelerating adoption and resulting in exponential-like growth during the early stage. However, as $S(t)$ approaches to m , saturation is reached, causing the growth to slow down and resemble logistic growth in the later stage. To solve this equation, I started by substituting the variable:

Let

$$x = \frac{S(t)}{m} \quad \Rightarrow \quad S(t) = mx$$

After differentiating both sides, we substitute it into the original equation:

$$\begin{aligned} \frac{dS}{dt} &= m \cdot \frac{dx}{dt} \\ m \cdot \frac{dx}{dt} &= (p + qx)(1 - x) \cdot m \\ \frac{dx}{dt} &= (p + qx)(1 - x) \end{aligned}$$

Now we rearrange it using separable variable,

$$\frac{1}{(p+qx)(1-x)} dx = dt$$

To integrate the left-hand side, we need to apply partial fractions:

$$\frac{1}{(p+qx)(1-x)} = \frac{A}{p+qx} + \frac{B}{1-x}$$

Multiply both sides by $(p+qx)(1-x)$,

$$1 = A(1-x) + B(p+qx)$$

$$1 = A - Ax + Bp + Bqx = (A + Bp) + (-A + Bq)x$$

Therefore,

$$A + Bp = 1 \quad (1)$$

$$-A + Bq = 0 \quad (2)$$

$$(1) + (2) \Rightarrow Bp + Bq = 1 \Rightarrow B(p+q) = 1$$

Then,

$$B = \frac{1}{p+q} \quad \text{and} \quad A = \frac{q}{p+q}$$

$$\frac{1}{(p+qx)(1-x)} = \frac{q}{p+q} \cdot \frac{1}{p+qx} + \frac{1}{p+q} \cdot \frac{1}{1-x}$$

$$\int \left(\frac{q}{p+q} \cdot \frac{1}{p+qx} + \frac{1}{p+q} \cdot \frac{1}{1-x} \right) dx = \int dt$$

Now we can integrate each term:

$$\frac{q}{p+q} \cdot \frac{1}{q} \ln |p+qx| - \frac{1}{p+q} \ln |1-x| = t + C$$

$$\frac{1}{p+q} (\ln |p+qx| - \ln |1-x|) = t + C$$

$$\ln \left(\frac{p+qx}{1-x} \right) = (p+q)(t+C)$$

$$\frac{p+qx}{1-x} = Ae^{(p+q)t}, \quad \text{where } A = e^{(p+q)C}$$

To solve for x ,

$$p+qx = Ae^{(p+q)t}(1-x)$$

$$p+qx = Ae^{(p+q)t} - Ae^{(p+q)t}x$$

$$qx + Ae^{(p+q)t}x = Ae^{(p+q)t} - p$$

$$x(q + Ae^{(p+q)t}) = Ae^{(p+q)t} - p$$

$$x = \frac{Ae^{(p+q)t} - p}{q + Ae^{(p+q)t}}$$

Since $S(t) = mx$,

$$S(t) = m \cdot \frac{Ae^{(p+q)t} - p}{q + Ae^{(p+q)t}}$$

To find A, we apply the condition $S(0) = 0$,

$$\begin{aligned} 0 &= m \cdot \frac{A - p}{q + A} \\ \Rightarrow A &= p \end{aligned}$$

Substitute $A = p$ back into $S(t)$,

$$S(t) = m \cdot \frac{pe^{(p+q)t} - p}{q + pe^{(p+q)t}} = m \cdot \frac{e^{(p+q)t} - 1}{\frac{q}{p} + e^{(p+q)t}}$$

Now we multiply both numerator and denominator by $e^{-(p+q)t}$ to simplify:

$$S(t) = m \cdot \frac{1 - e^{-(p+q)t}}{1 + \frac{q}{p}e^{-(p+q)t}}$$

To generate the Bass model for BOYNEXTDOOR's fan growth, I first needed to determine the value of innovation coefficient q and imitation coefficient p . The value of total market potential m was set to 15.0 million fans, which is consistent with the carrying capacity C used in the logistic model section.

As a long-time K-pop fan, I have realized that in the initial stages of any group's debut, there is often limited exposure through performances and variety shows. As a result, internal fans usually do not have enough content to promote the group through edited clips, trending hashtags, or any other social media engagement. Therefore, it is reasonable for me to assume that the fanbase growth in the first month following BOYNEXTDOOR's debut—specifically the 80,400 fans—was almost entirely the result of external influence, that is, innovation. With the value, I calculated the value of p using the following equation:

$$\begin{aligned} p &= \frac{S(1)}{m} \\ p &= \frac{80,400}{15,000,000} = 0.00536 \end{aligned}$$

Next, to estimate the value of q , I analyzed fan data from Month 2 to Month 20, under the assumption that the subsequent fan growth was shaped by both innovation and imitation. To calculate, I rearranged the discrete form of the Bass model equation:

$$q = \frac{\left(\frac{S_n - S_{n-1}}{m - S_{n-1}} \right) - p}{\frac{S_{n-1}}{m}}$$

After applying the formula across all months from Month 2 to Month 20, I obtained the individual values of q (refer to Appendix). To account for fluctuations in monthly data, I averaged the resulting values and this provided a final estimate of $q = 0.04175$. With these values, we can have our bass diffusion model for BOYNEXTDOOR's fan growth:

$$S(t) = \frac{15,000,000 \cdot (1 - e^{-(0.00536+0.04175)t})}{1 + \left(\frac{0.04175}{0.00536}\right) \cdot e^{-(0.00536+0.04175)t}}$$

$$S(t) = \frac{15,000,000 \cdot (1 - e^{-0.0471t})}{1 + 7.7894 \cdot e^{-0.0471t}}$$

Similarly, to assess the validity of my Bass Diffusion Model, I plotted the model curve alongside the actual data.

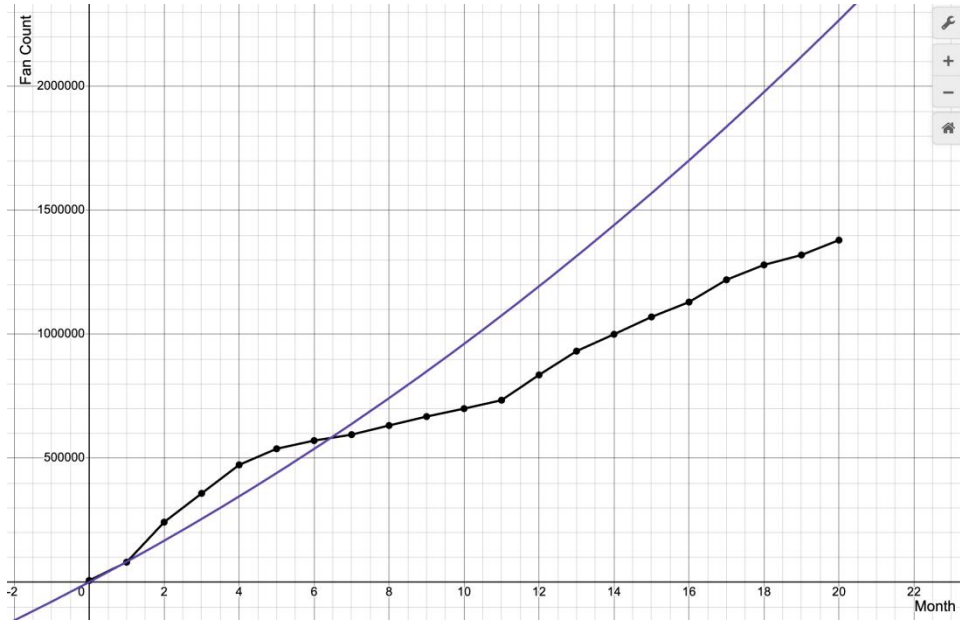


Figure 5: Comparison Between Actual Subscriber Data and the Bass Diffusion Model

As shown in the figure, compared to the previously explored population models, the Bass diffusion model provides a closer fit to real behavioral patterns. The rapid fan growth during BOYNEXTDOOR's debut is successfully captured, especially in the first month, which may be due to my choice of using the first month's fan count to estimate the value of innovator coefficient p . However, in the later stage, the model begins to overestimate the fan count, though the overall trend remained relatively close. This discrepancy is likely caused by estimation errors in the imitation coefficient q . During my estimation, I observed that calculating q sometimes yielded negative or incredibly small values, suggesting that the innovation coefficient may have been overestimated.

There are two key reasons behind this: first, I had to rely on monthly intervals due to data limitations, rather than more precise weekly fan count data; second, even during the debut period, imitation may have already contributed to fan acquisition rather than it being driven purely by innovation, even if only to a small extent.

Despite these estimation challenges, the Bass model is clearly more graphically consistent with the real-world fan behavior than the exponential or logical models. It is not constrained by the assumption of a constant growth rate, but instead account for the interaction of innovators and imitators. As it reflects not only the initial explosive growth but also the trend of saturation as the market potential has close, the model is both descriptive and predictive. Additionally, in term of the model itself, it offers interpretability, as it mirrors the dynamic structure of the K-pop industry.

Therefore, although the Bass model I obtained is not perfect, it demonstrates great potential. It is not only a mathematically satisfactory model, but also a practical one due to its incorporation of external marketing and social spread. Once the total market potential and coefficients p and q are determined, entertainment companies can apply this model to simulate fan growth for their groups and make informed decisions about resource allocation for upcoming comebacks. However, it's worth noting that these values remain difficult to determine precisely, as they are not constant and are affected by factors such as a group's evolving performance quality and virality of content.

An extension to the current Bass Diffusion Model would be to allow the coefficients p and q to vary over time, forming $p(t)$ and $q(t)$. During comeback periods, the innovation coefficient $p(t)$ may temporarily spike due to concentrated marketing, while $q(t)$ may gradually rise as fans have more performances to edit and share online. This time-varying coefficient model would be more adaptable to the behavioral patterns of K-pop fans and enable companies to capture the short-term surges in fan growth more accurately. In preparing for comebacks, such an extension would be especially helpful.

3. Comparison and Conclusion

The main aim of this investigation is to explore potential mathematical models that could effectively stimulate and predict the fan growth of K-pop group BOYNEXTDOOR. To achieve this, I studied four different models: polynomial regression, exponential growth model, logistic growth model, and the Bass Diffusion Model. Each model provided a unique perspective to fan growth, and varied in terms of its mathematical and real-world practicality.

As the starting point of my exploration, polynomial regression demonstrated strong numerical fitting ability. It produced a 7th-degree polynomial with an R^2 value of approximately 0.9994, successfully capturing small fluctuations in the monthly fan growth over the observed timeframe. However, it ultimately failed to achieve the broader goals of this study, as its purely data-driven nature meant it lacked interpretability for real-world fan behavior. Furthermore, its performance outside the observed range became highly unreliable, making it unable for predicting the trends of future fan growth.

The exponential model, though commonly used in growth modeling, performed poorly in this case as it failed to capture the initial surge in fan growth. The model's assumption of a constant growth rate R is unrealistic in the K-pop industry, where promotional efforts during each comeback period brings fluctuations to that. The later logistic model improved upon this by introducing a carrying capacity, allowing the curve to level off more realistically over time. However, the same assumption on R and its exponential-like behavior in the early stages made it similarly ineffective in modeling K-pop fan growth.

In contrast, the Bass Diffusion Model, by integrating both external innovation and internal imitation, avoided the limitations of assuming a constant growth rate. It successfully captured the initial explosive fans growth of BOYNEXTDOOR and offered a model that most closely resembles the fan growth patterns seen in the K-pop industry. Despite the challenges involved in estimating its parameters, this model displayed the greatest potential for modeling K-pop groups' fan expansion, as it uniquely combined interpretability, real-world relevance, and predictive value. For any idol group, the Bass Diffusion Model can serve as the most suitable forecasting tool and support their management team in making strategic decisions.

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5. Appendix

Month	S_n	S_{n-1}	ΔS	$\frac{S(n-1)}{m}$	q
2	242000	80400	161600	0.00536	1.02078
3	358000	242000	116000	0.01613	0.15497
4	473000	358000	115000	0.02387	0.10450
5	538000	473000	65000	0.03153	-0.02808
6	571000	538000	33000	0.03587	-0.08582
7	595000	571000	24000	0.03807	-0.09711
8	632000	595000	37000	0.03967	-0.07041
9	668000	632000	36000	0.04213	-0.06775
10	700000	668000	32000	0.04453	-0.07020
11	734000	700000	34000	0.04667	-0.06390
12	836000	734000	102000	0.04893	0.03658
13	932000	836000	96000	0.05573	0.02537
14	1000000	932000	68000	0.06213	-0.00905
15	1070000	1000000	70000	0.06667	-0.00540
16	1130000	1070000	60000	0.07133	-0.01475
17	1220000	1130000	90000	0.07533	0.01498
18	1280000	1220000	60000	0.08133	-0.01237
19	1320000	1280000	40000	0.08533	-0.02864
20	1380000	1320000	60000	0.08800	-0.01108